King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 243

Original exam date: 2 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The slope of the parametric curve

$$x = -4\cos\theta, \quad y = 4\sin\theta$$

at $\theta = \dfrac{\pi}{4}$ is

(a) 1

(b) -4

(c) 4

(d) $\dfrac{\sqrt{2}}{2}$

(e) -1

2. The length of the parametric curve

$$x = e^{-t}\cos t, \quad y = e^{-t}\sin t, \qquad 0 \le t \le 1$$

is equal to

(a) $1 - e^{-1}$

(b) $1 + e$

(c) $2\sqrt{2}\,e^{-1}$

(d) $\sqrt{2}\,(1 - e^{-1})$

(e) $1 + e^{-1}$

3. Let $\vec{u} = \langle a, b\rangle$ be the vector of magnitude 2 in the same direction as $\langle\sqrt{3}, 3\rangle$. Then $a + b^2 =$

(a) 3

(b) 1

(c) 4

(d) $\sqrt{3}$

(e) 2

4. A set of parametric equations of the line passing through $(-3, 1)$ and $(1, 9)$ is

(a) $x = 1 - t,\ y = 9 + 2t$

(b) $x = -3t,\ y = t$

(c) $x = -3 + 2t,\ y = 1 + t$

(d) $x = -3 + t,\ y = 1 + 2t$

(e) $x = 1 + 4t,\ y = 9 + t$

5. Let $\langle a, b, c\rangle$ be the vector projection of $\vec{u} = \langle 3, -5, 2\rangle$ onto $\vec{v} = \langle 7, 1, -2\rangle$. Then $a + b + c =$

(a) $\dfrac{4}{3}$

(b) $\dfrac{1}{9}$

(c) $\dfrac{14}{9}$

(d) $-\dfrac{5}{3}$

(e) $\dfrac{7}{9}$

6. The area of the surface formed by revolving the circle $r = 2\cos\theta$, $\theta \in [0, \pi]$ about the line $\theta = \dfrac{\pi}{2}$ is

(a) π^2

(b) $4\pi^2$

(c) π

(d) 4π

(e) 2π

7. The polar curve $r = 5\cos\theta$ is a circle with center $C(a, b)$ and radius c. Then $a + b + c =$

(a) 10

(b) $\dfrac{5}{2}$

(c) 5

(d) $\dfrac{15}{2}$

(e) 20

8. Which one of the following vectors is perpendicular to the tangent to the graph of $f(x) = \sqrt{25 - x^2}$ at the point $(4, 3)$?

(a) $\langle 3, 4\rangle$

(b) $\langle -4, 3\rangle$

(c) $\langle -3, 4\rangle$

(d) $\langle 4, 4\rangle$

(e) $\langle 4, 3\rangle$

9. A rectangular equation of the polar curve $\theta = \dfrac{3\pi}{4}$ is

(a) $y = -x$

(b) $y = -x + 1$

(c) $y = -\dfrac{\sqrt{2}}{2}x$

(d) $y = x$

(e) $y = \dfrac{\sqrt{2}}{2}x$

10. A vector orthogonal to both $\vec{u} = \langle -3, 2, -5\rangle$ and $\vec{v} = \langle 1, -1, 4\rangle$ is

(a) $\langle -3, -2, 1\rangle$

(b) $\langle 3, 7, 3\rangle$

(c) $\langle -4, 0, 1\rangle$

(d) $\langle 3, 7, 1\rangle$

(e) $\langle -1, 1, 1\rangle$

11. Consider the following three vectors

$$\vec{u} = \langle b, -5, 1\rangle, \quad \vec{v} = \langle 0, 1, -1\rangle, \quad \vec{w} = \langle 3, 1, 1\rangle.$$

If $\vec{u} \cdot (\vec{v} \times \vec{w}) = 36$, then $b =$

(a) 6

(b) 10

(c) 12

(d) 3

(e) 4

12. The parametric curve $x = \cos\theta + \theta\sin\theta,\ y = \sin\theta - \theta\cos\theta$ has

(a) a vertical tangent at $\theta = -\dfrac{\pi}{4}$

(b) a horizontal tangent at $\theta = \dfrac{\pi}{2}$

(c) a horizontal tangent at $\theta = \dfrac{3\pi}{2}$

(d) a horizontal tangent at $\theta = \pi$

(e) a vertical tangent at $\theta = \dfrac{\pi}{4}$

13. If $R(a, b, c)$ is the point that lies one-third of the way from $P(1, 2, 5)$ to $Q(6, 8, 2)$, then $3a+b+c =$

(a) $\dfrac{4}{3}$

(b) 8

(c) 16

(d) $\dfrac{8}{3}$

(e) 0

14. The area of the inner loop of $r = 1 + 2\sin\theta$ is

(a) $\int_{7\pi/6}^{3\pi/2}(1 + 2\sin\theta)^2\,d\theta$

(b) $\int_{\pi/6}^{\pi/2}(1 + 2\sin\theta)^2\,d\theta$

(c) $\dfrac{1}{2}\int_{0}^{2\pi}(1 + 2\sin\theta)^2\,d\theta$

(d) $\dfrac{1}{2}\int_{\pi/6}^{5\pi/6}(1 + 2\sin\theta)^2\,d\theta$

(e) $\dfrac{1}{2}\int_{-\pi/6}^{\pi/6}(1 + 2\sin\theta)^2\,d\theta$

15. A rectangular equation of the parametric curve

$$x = 2\cot\theta \quad \text{and} \quad y = 4\sin\theta\cos\theta, \qquad 0 < \theta < \pi$$

is

(a) $y(4 - x^2) = -8x$

(b) $4 + y^2 = x^2$

(c) $4 + x^2 = y^2$

(d) $y(4 + x^2) = 8x$

(e) $y(4 + x^2) = 16x$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\frac{dy}{dx} = \frac{4\cos\theta}{4\sin\theta} = \cot\theta = 1$ at $\theta = \frac{\pi}{4}$.
2	**(d)**	$(x')^2 + (y')^2 = 2e^{-2t}$, so $L = \sqrt{2}\int_0^1 e^{-t}dt = \sqrt{2}(1 - e^{-1})$.
3	**(c)**	$\vec{u} = 2\frac{\langle\sqrt{3},3\rangle}{2\sqrt{3}} = \langle 1, \sqrt{3}\rangle$, so $a + b^2 = 4$.
4	**(d)**	Direction $(4, 8) \parallel (1, 2)$ through $(-3, 1)$: $x = -3 + t,\ y = 1 + 2t$.
5	**(a)**	$\frac{u\cdot v}{v\cdot v} = \frac{12}{54} = \frac{2}{9}$; sum of components of $\frac{2}{9}v$ is $\frac{12}{9} = \frac{4}{3}$.
6	**(b)**	$S = 2\pi\int_0^\pi r\cos\theta\sqrt{r^2 + r'^2}\,d\theta = 8\pi\int_0^\pi \cos^2\theta\,d\theta = 4\pi^2$.
7	**(c)**	$x^2 + y^2 = 5x$: center $(\frac{5}{2}, 0)$, radius $\frac{5}{2}$, so $a + b + c = 5$.
8	**(e)**	Tangent direction $\langle 3, -4\rangle$ (slope $-\frac{4}{3}$); only $\langle 4, 3\rangle$ has dot product 0.
9	**(a)**	$\frac{y}{x} = \tan\frac{3\pi}{4} = -1$, so $y = -x$.
10	**(d)**	$\vec{u}\times\vec{v} = \langle 3, 7, 1\rangle$, and both dot products with $\vec{u}, \vec{v}$ are 0.
11	**(c)**	$\vec{v}\times\vec{w} = \langle 2, -3, -3\rangle$, so $2b + 12 = 36$ and $b = 12$.
12	**(d)**	$x' = \theta\cos\theta$, $y' = \theta\sin\theta$: at $\theta = \pi$, $y' = 0$, $x' \neq 0$, horizontal.
13	**(c)**	$R = P + \frac{1}{3}\overrightarrow{PQ} = (\frac{8}{3}, 4, 4)$, so $3a + b + c = 16$.
14	**(a)**	Inner loop $\frac{7\pi}{6} \le \theta \le \frac{11\pi}{6}$; by symmetry $A = \int_{7\pi/6}^{3\pi/2} r^2\,d\theta$.
15	**(d)**	$\sin\theta\cos\theta = \frac{\cot\theta}{1+\cot^2\theta}$ gives $y = \frac{8x}{4+x^2}$, i.e. $y(4 + x^2) = 8x$.